# Governance Under Adversarial Strategy Injection in Sequential Commons

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**Abstract**

We study governance under adversarial strategy injection in sequential commons through two linked studies. Study 1 uses a minimal fishery commons to compare central governance signals under repeated strategy turnover and held-out regime shifts. A matched mutation-versus-live-LLM baseline shows that monitoring with sanctions reduces held-out collapse relative to no governance for both injector types, and a broader top-down sweep over partner mixes and pressure levels identifies adaptive quotas as the strongest overall signal, with temporary closures remaining competitive in hostile partner mixes. Study 2 introduces Harvest Commons to compare governance architectures rather than only signal choices. Across a tiered matrix over social compositions, top-down-only, bottom-up-only, and hybrid architectures, the hybrid condition wins most cells on patch health and local overharvest control, though it often pays a welfare cost in harsher adversarial settings. The combined result is a two-level evaluation protocol: first identify effective central governance signals in a controlled commons, then test how broader governance architectures behave in a second substrate.

## 1 Introduction

Classic commons theory predicts over-extraction under open access and highlights institutional design as the key lever for sustainable use [2, 7, 11]. In machine learning, sequential social dilemma research has made the same point in a different language: cooperation is a property of policies interacting over time in stateful environments, so evaluation must account for long horizons, partial observability, and held-out social situations rather than one-shot payoffs [1, 3–5]. Recent LLM-agent commons work shows that cooperative behavior can emerge in simulated societies, but those findings remain fragile under changes in prompts, partner populations, and environmental conditions [9, 10]. What remains weakly characterized is whether governance itself stays effective once new exploitative strategies keep entering the population.

This paper studies that problem as *governance under adversarial strategy injection*. The paper-level question is broader than a single fishery result: *how can governance stabilize cooperation under repeated strategic turnover in sequential commons, and what changes when governance moves from fixed central signals to broader governance architectures?* To answer that question cleanly, the paper is split into two linked studies. Study 1 is an identification study in a minimal fishery commons. It asks which *central governance signals* remain effective under invasion pressure, first in a matched mutation-versus-live-LLM baseline and then in a broader top-down sweep over monitoring with sanctions, adaptive quotas, and temporary closures. Study 2 is an extension study in a second substrate, Harvest Commons. It asks a different downstream question: once central signals have been identified, how should governance itself be organized—purely top-down, bottom-up through local communication and transfers, or hybrid?

That split is deliberate. Commons research has long shown that communication, collective choice, and locally enforced rules can materially change outcomes even without full central command [8, 12]. At the same time, benchmark work in multi-agent learning increasingly argues that robust conclusions should survive transfer across substrates and social situations, not only one carefully tuned environment [3, 5]. The fishery study therefore isolates the signal-design question, while the Harvest study widens the lens to governance architecture. Together they form one paper because they address the same overarching problem from two adjacent levels: which central interventions work, and what changes when governance is no longer only a hard-coded central rule.

The contribution is therefore not "a new fishery environment" or "another LLM society demo." The contribution is a two-level invasion-stability protocol for sequential commons. First, we use repeated adversarial strategy turnover, held-out regime evaluation, and run-level uncertainty to compare central governance signals in a minimal renewable commons. Second, we transfer the pressure-test to a new commons substrate and compare top-down, bottom-up, and hybrid governance architectures. The resulting evidence is specific enough to support concrete claims, but broad enough to speak to decentralized AI settings where governance is part of the system design rather than an afterthought.

# 2 Methods

Both studies use the same core invasion protocol. A population of strategies is evaluated over generations, weaker strategies are removed, and new strategies are injected through a specified generator. Selection is performed in train conditions and reported robustness is measured in held-out test conditions. This separates adaptation pressure from generalization assessment and makes governance comparisons about out-of-distribution stability rather than only in-sample performance.

## 2.1 Study 1: Fishery Commons and central governance signals

The first study uses a minimal renewable fishery commons with a shared stock, per-agent harvest actions, regeneration, and collapse dynamics. Agents follow interpretable threshold strategies, and governance changes extraction pressure directly through monitoring, quota clipping, sanctions, adaptive quotas, or temporary closures. The environment logs requested harvest, realized harvest, quota-clipped harvest, repeat-offender rate, closure activity, and stock-recovery lag so that governance can be analyzed mechanistically rather than only through final collapse outcomes.

Study 1 has two layers. The first is a matched baseline that compares `none`, `monitoring`, and `monitoring_sanctions` across two injector types: mutation and live JSON-based LLM injection via Ollama using `qwen2.5:3b-instruct`. This baseline is run on the easy_v1 and medium_v1 tiers. The second layer is a broader top-down sweep, restricted to medium_v1, that compares `none`, `monitoring_sanctions`, `adaptive_quota`, and `temporary_closure` under three non-LLM injector families: mutation, an adversarial heuristic injector, and search over mutations. The sweep also varies partner composition (`balanced` and `adversarial_heavy`) and adversarial pressure (0.3 and 0.5). This second layer is what identifies which central governance signals remain effective once the setting becomes more hostile.

Table 1: Shared run settings for the matched Fishery Commons baseline.

| Setting | Value |
|---|---|
| Generations | 15 |
| Train seeds per generation | 32 |
| Test seeds per generation | 32 |
| Replacement fraction | 0.2 |
| Adversarial pressure | 0.3 |
| Train regeneration rate | 2.0 |
| Train observation noise std | 6.0 |

### 2.2 Study 2: Harvest Commons and governance architecture

The second study introduces Harvest Commons, a second renewable commons substrate built around local patches rather than a single global stock. Agents remain self-interested, but the substrate adds local communication, limited credit transfers, and targeted intervention through a government agent. The purpose is not to repeat the same top-down question in another skin. The purpose is to ask what changes when governance is treated as an architecture choice.

Study 2 compares four governance conditions: `none`, `top_down_only`, `bottom_up_only`, and `hybrid`. Top-down-only uses targeted caps from the government agent. Bottom-up-only uses local communication and side-payments without central intervention. Hybrid combines the two. The matrix spans three tier presets (`easy_h1`, `medium_h1`, and `hard_h1`) and three social mixes (`cooperative_heavy`, `mixed_pressure`, and `adversarial_heavy`) with 12 runs per cell. In addition to patch health and welfare, the study reports local mechanism metrics including local aggression and neighborhood overharvest. These are needed because a governance architecture can change *where* exploitation happens, not only the global average outcome.

### 2.3 Statistical reporting

The primary Study 1 matched-baseline outcome is held-out collapse reduction, reported as the difference between no governance and monitoring with sanctions. Supporting metrics are held-out stock, time-to-collapse, and per-regime survival. Confidence intervals for the matched baseline are run-level bootstrap intervals with 5,000 resamples. The broader Study 1 signal sweep uses the same run summaries but is interpreted as a ranking study: adaptive quota, temporary closure, and monitoring with sanctions are compared on collapse first and welfare second.

The primary Study 2 comparison is the paired difference `hybrid`$-$`top_down_only` at matched run identifiers within each tier and social-mix cell. This comparison isolates whether local communication and credit-sharing add anything once central intervention is already present. Confidence intervals are again estimated by bootstrap over paired deltas.

Formal invasion dynamics, the definition of collapse robustness, and the saturation identifiability condition are given in Section 3. In the empirical pipeline, these quantities are estimated from run-level summaries and frozen result artifacts to avoid post-hoc threshold drift.

## 3 Formal Invasion-Stability Model

This section formalizes the invasion process evaluated in the experiments. Let $\mathcal{S}_t$ denote the strategy pool at generation $t$, with population distribution $p_t \in \Delta(\mathcal{S}_t)$. Let $g \in \mathcal{G}$ denote governance condition,

and let $r \sim R_{\text{train}}$ or $r \sim R_{\text{test}}$ denote train or held-out regime draws. In Study 1, $g$ indexes central signals such as monitoring with sanctions, adaptive quotas, and temporary closures. In Study 2, $g$ indexes governance architectures such as top-down-only, bottom-up-only, and hybrid control.

For strategy $s \in \mathcal{S}_t$, define train-regime fitness as

$$F_t(s; g) = \mathbb{E}_{r \sim R_{\text{train}}} \left[ U(s, p_t, g, r) \right],$$

where $U$ is episodic return. Selection keeps a renormalized top-mass distribution $\mathcal{T}_q(p_t, F_t)$, where $q \in (0, 1)$ is replacement fraction. New mass is injected by generator $\mathcal{I}_i$, where $i$ indexes injector type (mutation or live LLM). The population update is

$$p_{t+1} = (1 - q)\, \mathcal{T}_q(p_t, F_t) + q\, \mathcal{I}_i(\mathcal{H}_t),$$

with history $\mathcal{H}_t$ containing observed strategy performance up to generation $t$.

Let

$$C(p_t, g, r) = \mathbb{E}[\mathbf{1}\{\text{collapse}\} \mid p_t, g, r]$$

be collapse probability at generation $t$. For a horizon $T$, define invasion-robustness on collapse under governance $g$ versus no governance as

$$\mathcal{R}_{g,i}^{\text{collapse}}(T) = \frac{1}{T} \sum_{t=1}^{T} \mathbb{E}_{r \sim R_{\text{test}}} \left[ C\left(p_t^{\text{none},i}, \text{none}, r\right) - C\left(p_t^{g,i}, g, r\right) \right].$$

Positive values indicate lower collapse under governance during invasion. The paper's reported $\Delta_{g,i}^{\text{collapse}}$ is the empirical estimate of this quantity.

A practical inference rule follows directly: governance is considered invasion-stable on collapse when the lower bound of the 95% confidence interval for $\mathcal{R}_{g,i}^{\text{collapse}}(T)$ is strictly positive.

**Proposition (Saturation non-identifiability).** If for a regime pack $R$, horizon $T$, and all governance conditions $g \in \mathcal{G}$,

$$C\left(p_t^{g,i}, g, r\right) = 1 \quad \forall t \in \{1, \ldots, T\},\ \forall r \sim R,$$

then collapse-based governance ranking is non-identifiable because $\mathcal{R}_{g,i}^{\text{collapse}}(T) = 0$ for all $g$.

*Sketch.* Substituting $C = 1$ into the definition of $\mathcal{R}_{g,i}^{\text{collapse}}(T)$ yields zero difference at every time and regime, so all governance conditions tie under collapse-based scoring.

This proposition explains why hard-tier saturation is treated as stress confirmation rather than mechanism-separation evidence.

# 4 Results

## 4.1 Study 1: central governance signals in Fishery Commons

The matched Fishery Commons baseline shows that even a minimal central intervention survives repeated injection pressure. Under five paired high-power runs in medium_v1, monitoring with sanctions reduces held-out collapse relative to no governance for both injector types. For mutation, collapse reduction is 0.1099 with 95% CI $[0.0310, 0.1896]$. For live JSON-based LLM injection, collapse reduction is 0.0835 with 95% CI $[0.0154, 0.1509]$. Mean held-out stock also rises substantially, by 23.35 for mutation and 22.40 for live LLM injection. These matched results establish that the invasion-stability protocol is not an artifact of one injector family and that a lightweight central signal already improves robustness.

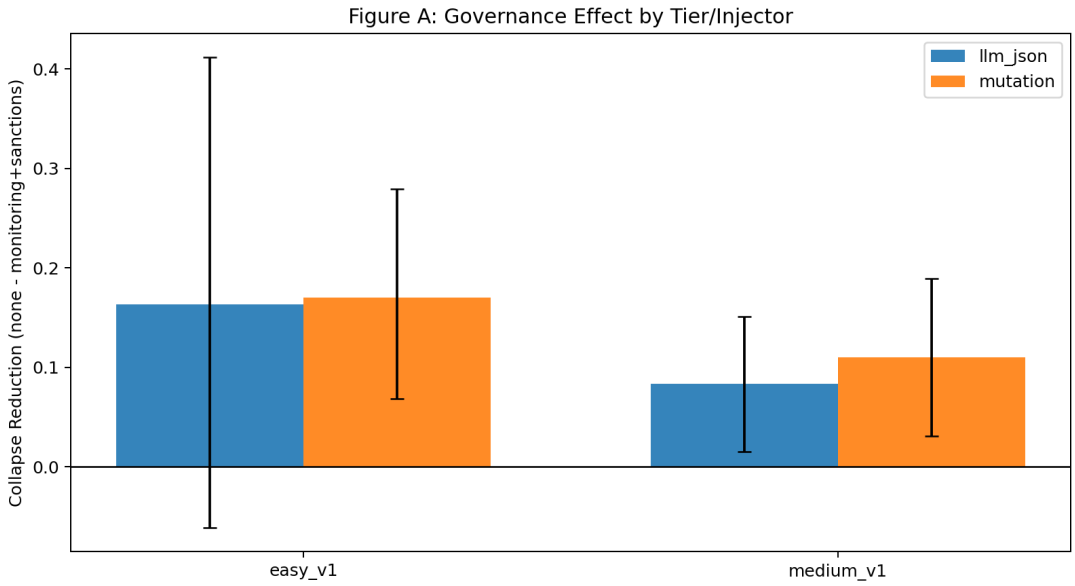


Figure 1: Matched Fishery Commons baseline. Monitoring with sanctions reduces held-out collapse relative to no governance for both mutation and live LLM injection. Error bars show 95% confidence intervals.

The broader top-down sweep in medium_v1 sharpens the central-governance question. Across balanced and adversarial-heavy partner mixes, pressure levels 0.3 and 0.5, and three non-LLM injector families, `adaptive_quota` is the most frequent winner in every injector block. It wins all four scenarios for mutation, all four for the adversarial heuristic injector, and all four for search over mutations. This happens because `adaptive_quota` and `temporary_closure` tie on collapse reduction in most cells, but adaptive quotas usually retain more welfare outside the most hostile partner mixes.

Table 2: Study 1b top-down signal summary in medium_v1. Collapse reductions are relative to no governance; welfare difference is `adaptive_quota` minus `temporary_closure`.

| Injector | Sanctions | Adaptive | Closure | Adaptive - closure welfare |
|---|---|---|---|---|
| Mutation | 0.3737 | 0.9167 | 0.9167 | 2408.81 |
| Adversarial heuristic | 0.0208 | 0.9167 | 0.9167 | 2211.61 |
| Search over mutations | 0.2617 | 0.9167 | 0.9167 | 2723.02 |

The nuance is important. In balanced partner mixes, adaptive quotas dominate closures on welfare by large margins: 4792.55 for mutation, 5202.70 for the adversarial heuristic injector, and 5367.27 for search over mutations at pressure 0.3. In adversarial-heavy partner mixes, that advantage shrinks or reverses: the same adaptive-minus-closure welfare difference is −215.62, −779.48, and −71.33 at pressure 0.3 for the three injectors. So the correct interpretation is not that temporary closures fail. It is that adaptive quotas are the strongest overall signal, while closures remain a serious hostile-population fallback when ecological protection is prioritized over welfare retention.

Mechanism logging explains why monitoring with sanctions is weaker than the more forceful top-down alternatives. Mean quota-clipped harvest under monitoring with sanctions is 8.53 for mutation, 18.32 for the adversarial heuristic injector, and 9.61 for search over mutations, while mean

repeat-offender rates remain 0.3032, 0.5568, and 0.3458. Monitoring with sanctions therefore reduces harm without fully suppressing aggressive extraction, whereas adaptive quotas and temporary closures push collapse to zero in nearly every Study 1b cell.

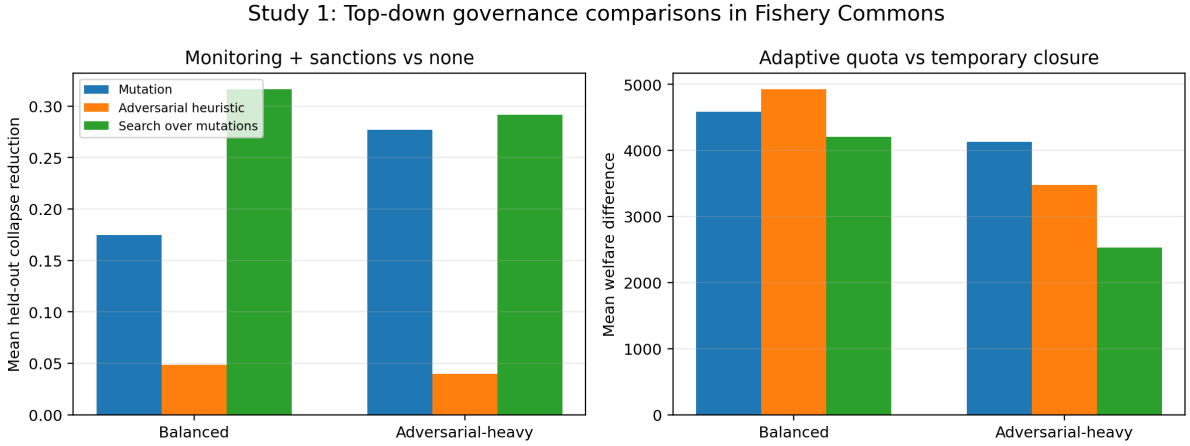


Figure 2: Study 1b summary. Left: mean held-out collapse reduction from monitoring with sanctions relative to no governance. Right: mean welfare advantage of adaptive quotas over temporary closures. Adaptive quotas dominate in balanced settings but lose some advantage in adversarial-heavy mixes.

### 4.2 Study 2: governance architecture in Harvest Commons

Study 2 shifts from signal design to governance architecture. Across the full Harvest Commons matrix of three tiers, three social mixes, four governance conditions, and 12 runs per cell, `hybrid` is the best condition in 7/9 tier-by-mix cells. `top_down_only` wins one cell and `bottom_up_only` wins one cell. Averaged over the full matrix, the paired difference `hybrid` − `top_down_only` is +0.1954 for patch health and −0.2431 for neighborhood overharvest, but −0.3748 for welfare. The high-level pattern is therefore consistent: hybrid governance usually controls local ecological damage better than central control alone, but often does so by accepting some welfare cost.

Table 3: Study 2 Harvest Commons summary from the full matrix. Positive deltas mean `hybrid` exceeds `top_down_only`; negative values are improvements for lower-is-better metrics such as neighborhood overharvest.

| Summary quantity | Value |
|---|---|
| Hybrid best-condition cells | 7 / 9 |
| Top-down-only best-condition cells | 1 / 9 |
| Bottom-up-only best-condition cells | 1 / 9 |
| Mean patch-health delta | +0.1954 |
| Mean welfare delta | -0.3748 |
| Mean local-aggression delta | -0.0037 |
| Mean neighborhood-overharvest delta | -0.2431 |

The paired bootstrap intervals show where that trade-off is strongest. In `medium_h1` with `adversarial_heavy` social mix, the patch-health delta is +0.0460 with 95% CI $[0.0250, 0.0650]$, while the welfare delta is −0.7439 with 95% CI $[-0.9683, -0.5662]$. In `hard_h1` with `adversarial_heavy`, the corresponding deltas are +0.0677 with 95% CI $[0.0325, 0.1025]$ and −0.3722 with 95% CI

$[-0.7746, -0.0330]$. By contrast, `hard_h1` with `mixed_pressure` is closer to a balanced trade-off: patch health improves by $+0.0552$ with 95% CI $[0.0070, 0.1016]$, while the welfare interval $[-0.0121, 0.3617]$ overlaps zero. This is the key Study 2 result: hybrid governance is not universally better, but it is consistently better on local ecological control and only sometimes better on welfare.

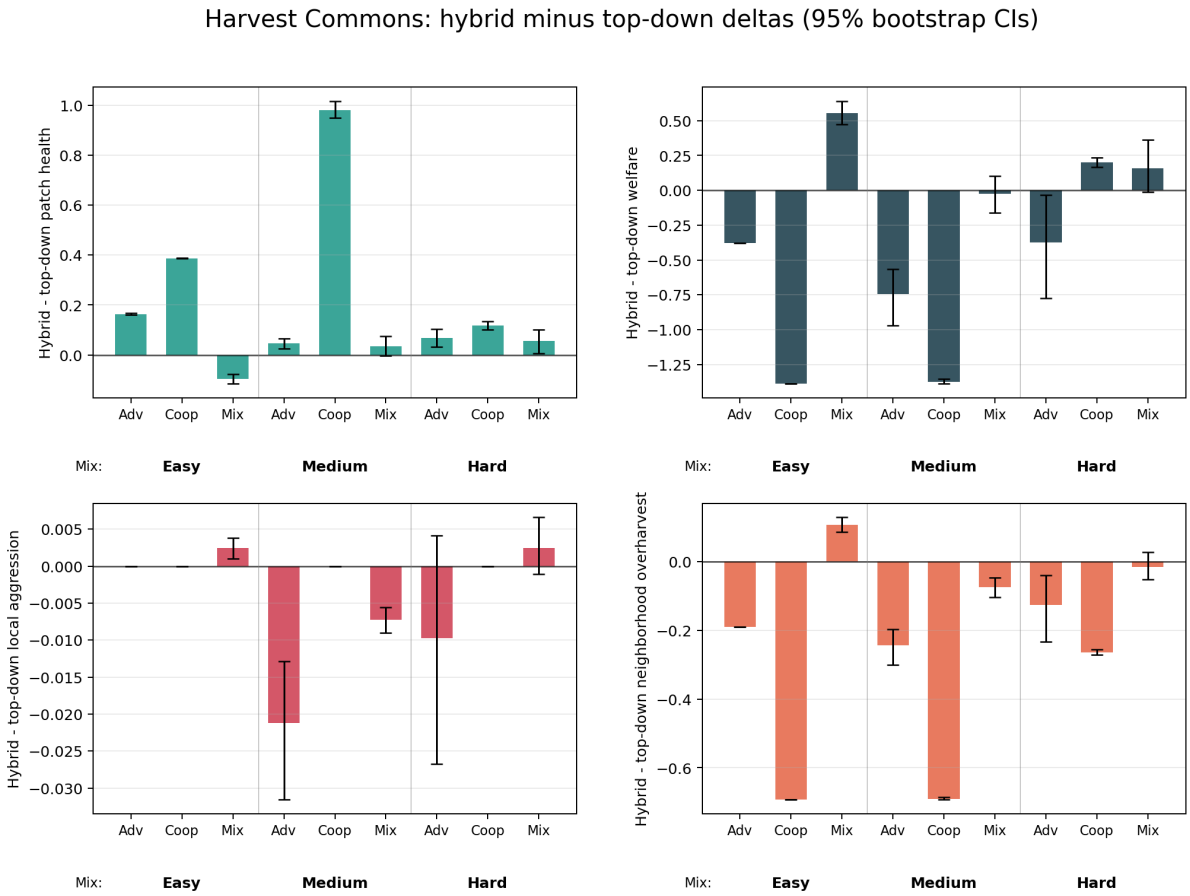


Figure 3: Study 2 paired `hybrid` minus `top_down_only` deltas in Harvest Commons with 95% bootstrap confidence intervals. Hybrid governance tends to improve patch health and reduce neighborhood overharvest, but its welfare effect depends strongly on tier and social mix.

Together, the two studies give a layered result. In the controlled fishery setting, central governance signals can be ranked and adaptive quotas emerge as the strongest overall top-down signal under invasion pressure. In the second substrate, once governance is allowed to become architectural rather than only signal-level, hybrid governance often produces the best local ecological control, but with a welfare trade-off that depends on social composition and task difficulty.

## 5 Discussion

The paper now supports two related claims rather than one narrow claim. First, central governance signals can be ranked under adversarial strategy injection in a minimal sequential commons. Study 1 shows that monitoring with sanctions is already enough to improve held-out collapse under both mutation and live LLM injection, which is the minimal robustness result. The broader signal sweep then shows that stronger top-down interventions are not interchangeable: adaptive quotas are

the best overall policy in medium-tier fishery runs, while temporary closures remain a credible fallback under adversarial-heavy partner mixes. This is a more useful conclusion than simply saying "governance helps," because it distinguishes light-touch enforcement from stronger adaptive intervention.

Second, governance should not be treated only as a central toggle. Study 2 asks what happens when the governance question is widened from signal design to architecture design. The answer is that hybrid governance can improve local ecological control beyond top-down-only control, but not for free. In Harvest Commons, hybrid governance usually wins on patch health and neighborhood overharvest, yet often sacrifices welfare in harsher adversarial settings. This is exactly the kind of trade-off that disappears if evaluation only reports a single collapse number. The Harvest results therefore justify the move from a single global stock to a second substrate with local interactions and communication.

That broader framing is why the two studies belong in one paper. Study 1 isolates which central signals work in a highly controlled environment. Study 2 then asks whether governance remains purely central once local coordination, communication, and targeted intervention are allowed to matter. Commons experiments have long suggested that communication and collective choice can improve outcomes even without a single command-and-control institution [8, 12]. The present results update that intuition for adversarially changing agent populations: once strategy turnover is explicit, governance must be evaluated both as a set of central signals and as a broader architecture for decentralized systems.

This also clarifies why people should care beyond fisheries or orchards. The contribution is not fish-specific. It is about institution design for self-interested multi-agent systems under uncertainty and turnover. The same protocol can be read as a testbed for shared compute budgets, energy systems, congestion control, spectrum allocation, or decentralized markets, where new behaviors keep entering and governance must balance resilience against performance. Recent generative-society work argues for realistic evaluation under diverse partners and conditions [6, 9, 10]; the present paper adds a more specific lesson that governance should be tested both at the level of signal choice and at the level of governance architecture.

## 6 Limitations

The two studies answer adjacent questions, not identical ones. Study 1 is a controlled signal-ranking study in Fishery Commons, while Study 2 is a governance-architecture extension in Harvest Commons. That makes the paper intellectually coherent, but it also means Study 2 should not be read as a strict replication of Study 1 in another substrate.

The LLM comparison currently appears only in Study 1. This is enough to show that minimal central governance survives both mutation and live JSON-based strategy generation, but it does not yet establish the same result for Study 2. Harvest Commons still uses structured non-LLM agents with communication and side-payments rather than live language-model control.

Hard settings remain useful mainly as stress evidence. In fishery, once collapse saturates at 1.0 and survival at 0.0 across conditions, governance separation becomes non-identifiable under floor effects. In Harvest Commons, the harder tiers remain informative, but welfare trade-offs become sharper and require more runs if one wants stronger cell-by-cell claims.

Finally, both substrates remain intentionally stylized. That is a strength for identification, but it limits external realism. Market feedbacks, richer heterogeneity, richer language interaction, and explicit institutional bargaining are natural next additions once the current governance comparisons are frozen.

# 7 Appendix

The appendix-only material serves two roles. For Study 1, hard-tier fishery runs remain useful as stress confirmation but are not used for causal ranking once collapse saturates across all governance conditions. For Study 2, the full tier-by-social-mix Harvest matrix, paired delta tables, and confidence-interval summaries are maintained as supplementary artifacts so that every number in the text can be traced to a frozen output file.

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